GEBBY YOLANDA NURHALIZA

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PROFESSIONAL SUMMARY

Software & Automation Engineer and Instrumentation Physics graduate with 3+ years of experience developing web applications, AI document tools, and IoT monitoring systems. Experienced in building and maintaining internal company applications using PHP (CodeIgniter 3), React.js, Next.js, PostgreSQL, and MySQL. Skilled in AI integration (LLM, RAG, OCR), database optimization, real-time monitoring dashboards, and solar-powered IoT devices.

EDUCATION

Universitas Muhammadiyah Riau – Riau, Indonesia
Bachelor Degree in Instrumentation Physics
Graduated: 2022 | GPA: 3.67/4.00

PROFESSIONAL EXPERIENCE

Software & Automation Engineer || Web Development – PT Riau Sakti United Plantations

Pulau Burung, Riau, Indonesia | Nov 2025 – Present

- Migrate legacy CodeIgniter 3 ERP modules to React.js and Next.js with TypeScript, including the accounting Purchase Invoice module covering list, search, edit, and automatic journal calculation.
- Build and maintain an AI Document Assistant (RAG-based) using PHP/CodeIgniter 3 and the Groq LLM API, with context-aware chat, OCR-based text extraction from PDF, image, and ZIP uploads, and structured field extraction (JSON schema) from unstructured documents.
- Develop auto-fill pipelines that save AI-extracted data (invoices, booking receipts, SP bills, meter readings) directly into the database, with automated template-based Excel reports.

Automation & IoT

- Design and build solar-powered IoT devices for the plantation, including coconut nursery irrigation, canal water level monitoring for flood and drought control, soil pH and moisture monitoring, and wind speed and direction alerts via Telegram.
- Built a Put To Light warehouse system using ESP32 and WebSocket to help staff locate products faster.

Software Engineer – PT Riau Sakti United Plantations

Pulau Burung, Riau, Indonesia | Mar 2023 – Nov 2025

- Developed and maintained the production department website for WTP Filling, WTP Processing, and DRP, covering input forms, daily data, production reports, and multi-level shift approval using CodeIgniter 3, PostgreSQL, Bootstrap, and JavaScript.
- Built monitoring systems for B/L code reports and customer cards, and maintained the ZHL Singapore accounting system, including payable and receivable transactions.
- Performed database integration, dynamic data manipulation, and report generation in PDF and Excel formats across production and accounting modules.
- Developed the front end of a real-time electrical distribution monitoring dashboard using React.js, covering kW, kVAR, kWh, and power factor per panel, 3-phase current and voltage, and per-second live charts via WebSocket.
- Implemented anomaly alerts, PDF and Excel consumption reports per department, and user management with role-based access control and OTP authentication.
- Collaborated with the back-end engineer to integrate REST APIs and WebSocket for real-time kWh data; the dashboard is live on the main PLTS UHT distribution panel.

PUBLICATION

Rancang Bangun Sistem Otomasi Pengatur Nutrisi pada Tanaman Hidroponik Nutrient Film Technique (NFT) dengan Sistem Pembangkit Listrik Off Grid Photovoltaic

Bachelor's Thesis | Universitas Muhammadiyah Riau

- Designed an automated nutrient control system for NFT hydroponic plants powered by an off-grid photovoltaic system.

ORGANIZATIONAL EXPERIENCE

Volunteer – Yayasan Bahtera Dwipa Abadi (YBDA) | 2025

- Volunteered in the Pompin "Pompong Pintar" program in Riau.

TECHNICAL SKILLS

- Programming Languages: PHP, JavaScript, TypeScript, C++
- Frameworks & Libraries: CodeIgniter 3, React.js, Next.js, Bootstrap
- Database: PostgreSQL, MySQL
- AI Integration: LLM API (Groq), RAG, OCR-based Document Processing
- IoT & Hardware: Arduino, ESP32, Sensor Integration, Solar Panel Installation
- Tools: Git, REST API, WebSocket, Matlab, Proteus, EWB, Fritzing, ArcGIS, SketchUp
- Other Skills: Problem Solving, Critical Thinking, Teamwork, Leadership, Time Management, Communication, Adaptability

LANGUAGES

Indonesian – Native
English – Intermediate